

Revolutions towards Embedded Systems & Cyber Physical Systems

Course Teacher:

Md. Obaidur Rahman, Ph.D.

Professor

Department of Computer Science and Engineering (CSE)
Dhaka University of Engineering & Technology (DUET), Gazipur.

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Course Title: Peripherals, Interfacing and Embedded Systems
Department of Computer Science and Engineering (CSE),
Islamic University of Technology (IUT), Gazipur.

Lecture References:

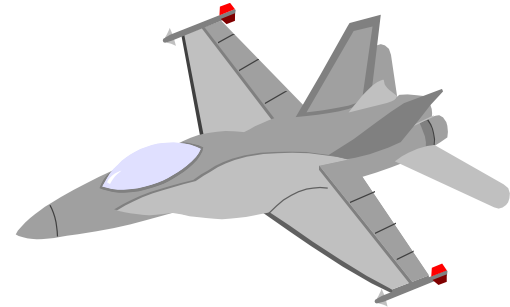
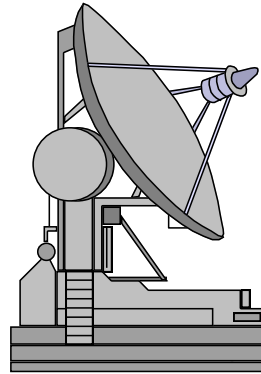
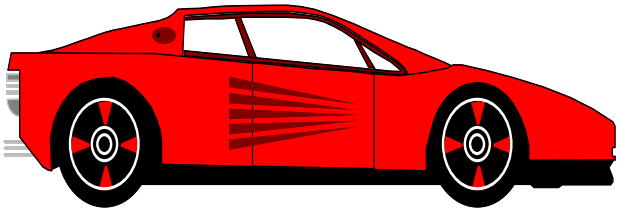
- ▶ **Book:**

- ▶ *Embedded System Design*, **Author:** P. Marwedel
- ▶ *Embedded System Design: An Introduction to Processes, Tools and Techniques*, **Author:** Arnold Berger, Arnold S. Berger

- ▶ **Lecture Materials:**

- ▶ <http://webct.edtec.unsw.edu.au/>
- ▶ *Saeid Nooshabadi*, Microprocessor and Interfacing

What is an Embedded System ??



Examples of Embedded Systems

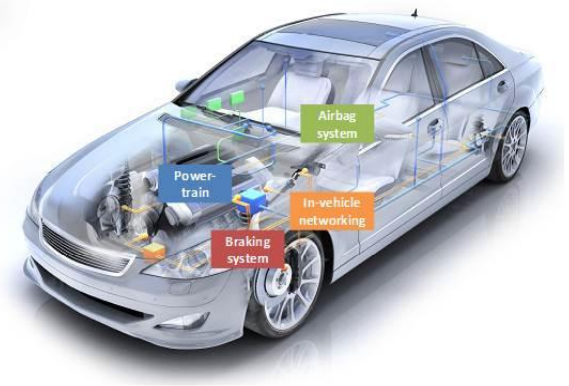
Office systems and mobile equipment	Building systems	Manufacturing and Process Control
Answering machines Copiers Faxes Laptops and notebooks Mobile Telephones PDAs, Personal organisers Still and video cameras Telephone systems Time recording systems Printer Microwave	Air conditioning Backup lighting and generators Building management systems CCTV systems Fire Control systems Heating and ventilating systems Lifts, elevators, escalators Lighting systems Security systems Security cameras Sprinkler systems	Automated factories Bottling plants Energy control systems Manufacturing plants Nuclear power stations Oil refineries and related storage facilities Power grid systems Power stations Robots Switching systems Water and sewage systems

Example of Embedded Systems

Transport	Communications	Other equipment
Aeroplanes Trains Buses Marine craft Jetties Automobiles Air Traffic Control Signalling Systems Radar Systems Traffic Lights Ticketing machines Speed cameras, Radar speed detectors	Telephone systems Cable systems Telephone switches Satellites Global Positioning System	Automated teller systems Credit card systems Medical Imaging equipment Domestic Central Heating control VCRs

Automotive Embedded Systems

- ▶ Today's high-end automobile may have 100 microprocessors:
 - ▶ 4-bit microcontroller checks seat belt;
 - ▶ Microcontrollers run dashboard devices;
 - ▶ 16/32-bit microprocessor controls engine.

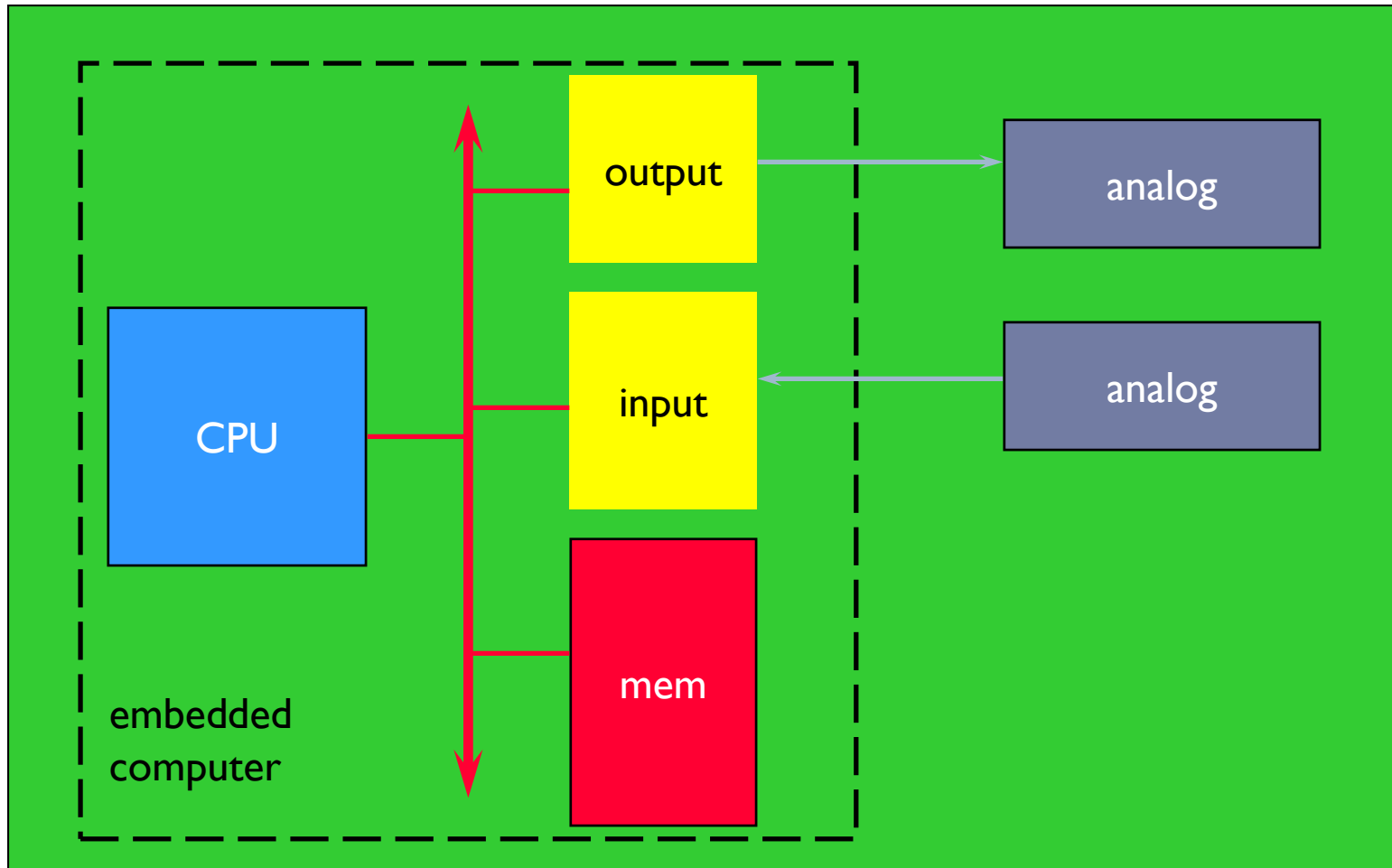


- ▶ **Example:**
- ▶ BMW 850i brake and stability control system
 - ▶ **Anti-lock brake system (ABS):** pumps brakes to reduce skidding.
 - ▶ **Automatic stability control (ASC+T):** controls engine to improve stability.
 - ▶ ABS and ASC+T communicate.
 - ▶ ABS was introduced first---needed to interface to existing ABS module.

Embedded Systems Rule the Market

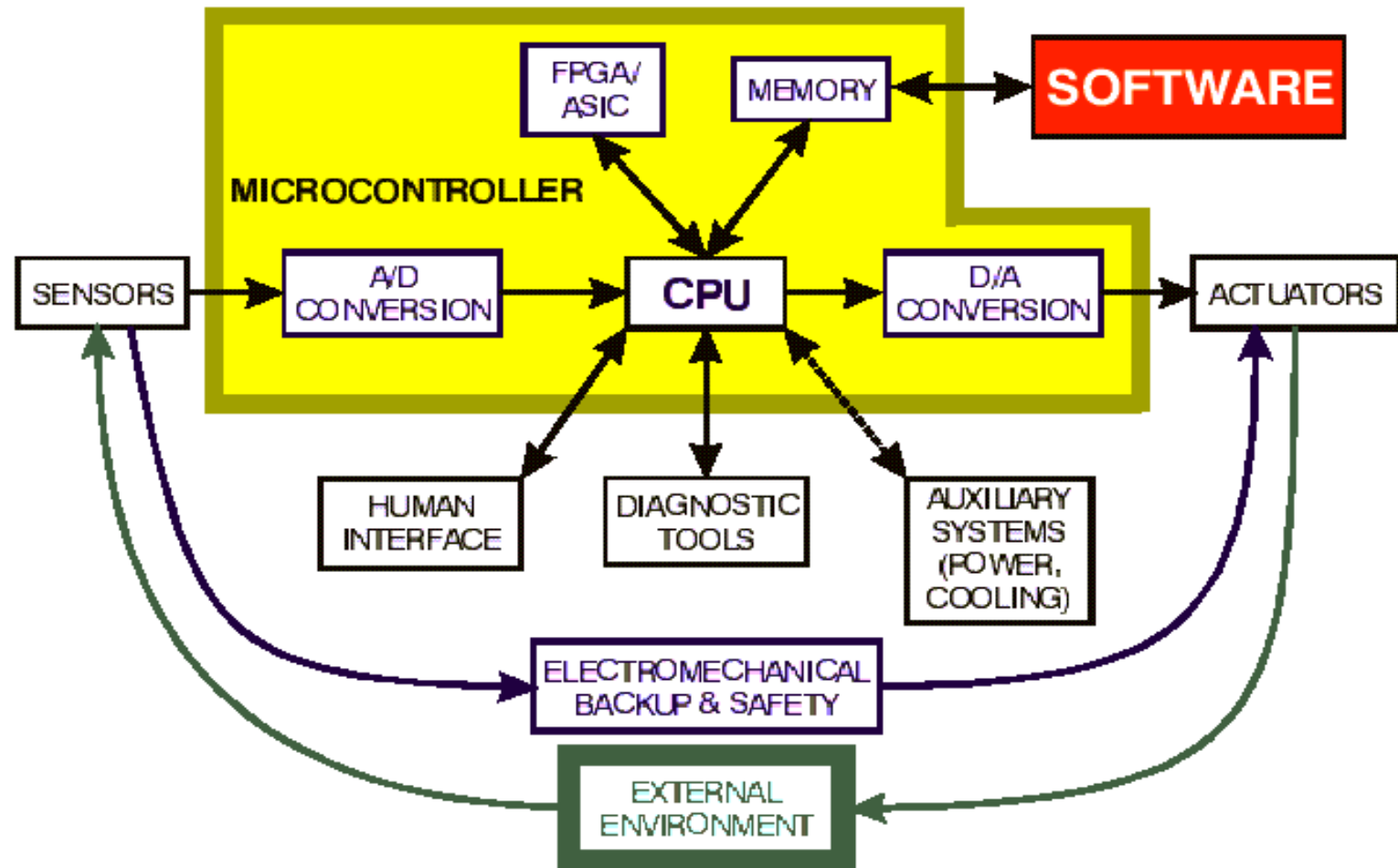
- ▶ **80 Million PCs vs. 3 Billion Embedded CPUs Annually**
- ▶ Embedded market is growing; whereas PC market is mostly saturated

Embedding a Computer



An Embedded Control System Designer's View

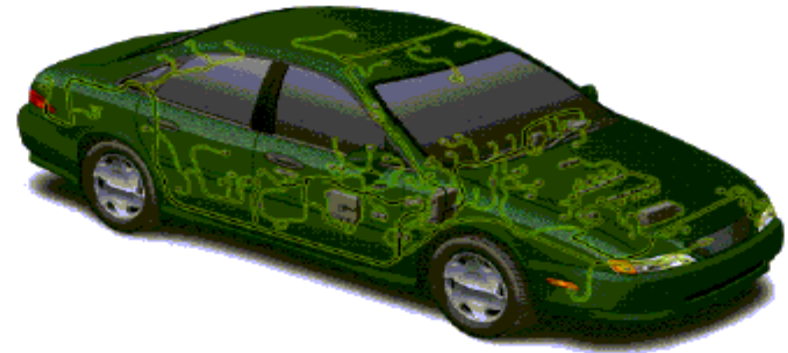
- ◆ Measured by: Cost, Time-to-market, Cost, Functionality, Cost & Cost.



A Customer View



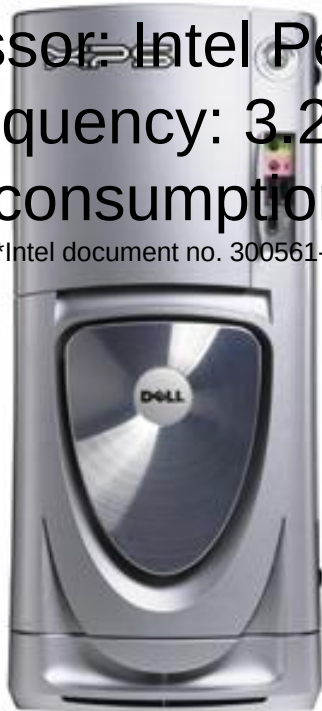
- ◆ **Reduced Cost**
- ◆ **Increased Functionality**
- ◆ **Improved Performance**
- ◆ **Increased Overall Dependability**
 - (Debatable, but can be true)



At the high end of the scale

Processor: Intel Pentium 4
Frequency: 3.2GHz
Power consumption: 103 W

*Intel document no. 300561-002



At the low end of the scale



Processor can consume no more
than 250-300mW

Embedded Devices

Computer Technology → Dramatic Change

▶ Processor

- ▶ 2X in speed every 1.5 years;
100X performance in last decade

▶ Memory

- ▶ DRAM capacity: 2X / 2 years; **64X size in last decade**
- ▶ Cost per bit: improves about 25% per year

▶ Disk

- ▶ capacity: > 2X in size every 1.0 years
- ▶ Cost per bit: improves about 100% per year
- ▶ **250X size in last decade**

Computer Technology → Dramatic Change!

- ▶ State-of-the-art PC when you graduate:

(at least...)

- ▶ Processor clock speed: 5000 MegaHertz
(5.0 GigaHertz)
- ▶ Memory capacity: 4000 MegaBytes
(8.0 GigaBytes)
- ▶ Disk capacity: 2000 GigaBytes
(2.0 TeraBytes)
- ▶ New units! Mega => Giga, Giga => Tera

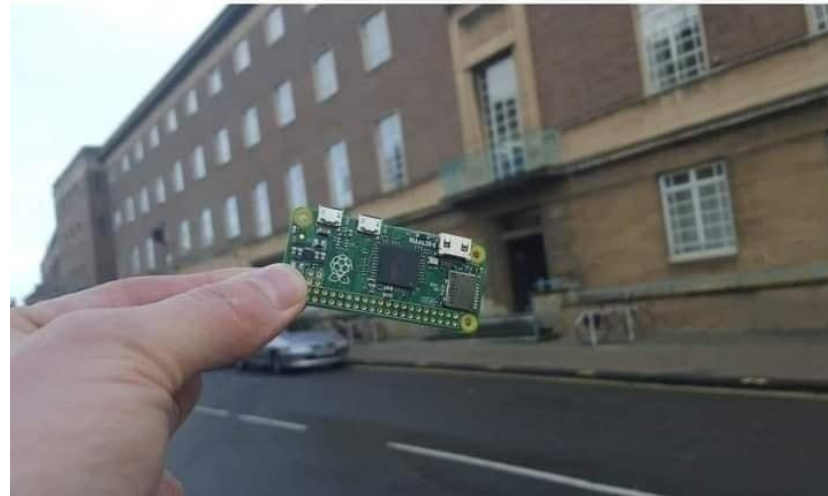
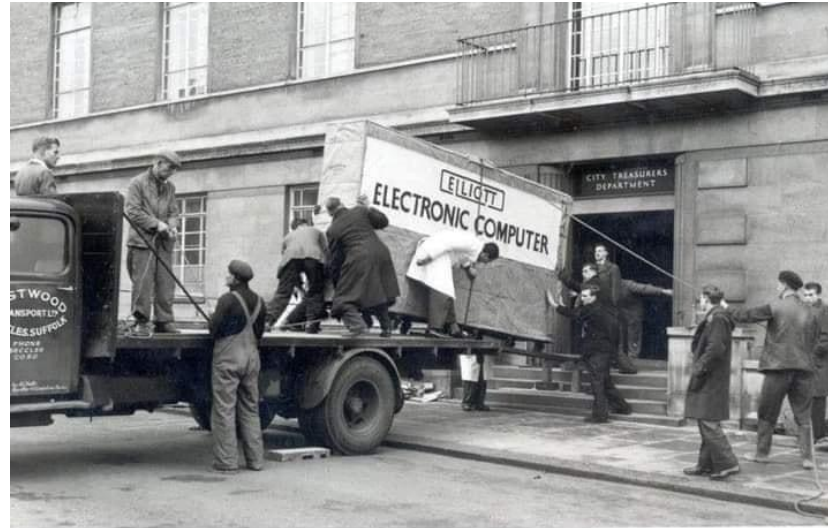
(Kilo, Mega, Giga, Tera, Peta, Exa, Zetta, Yotta = 10^{24})

Now a days, Kilo, Mega, etc. are incorrect Terminologies!

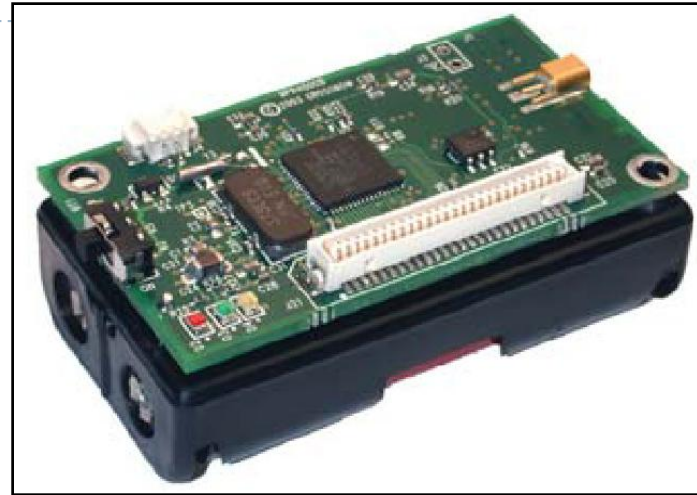
Computer Technology → Dramatic Change!



Computer Technology → Dramatic Change!

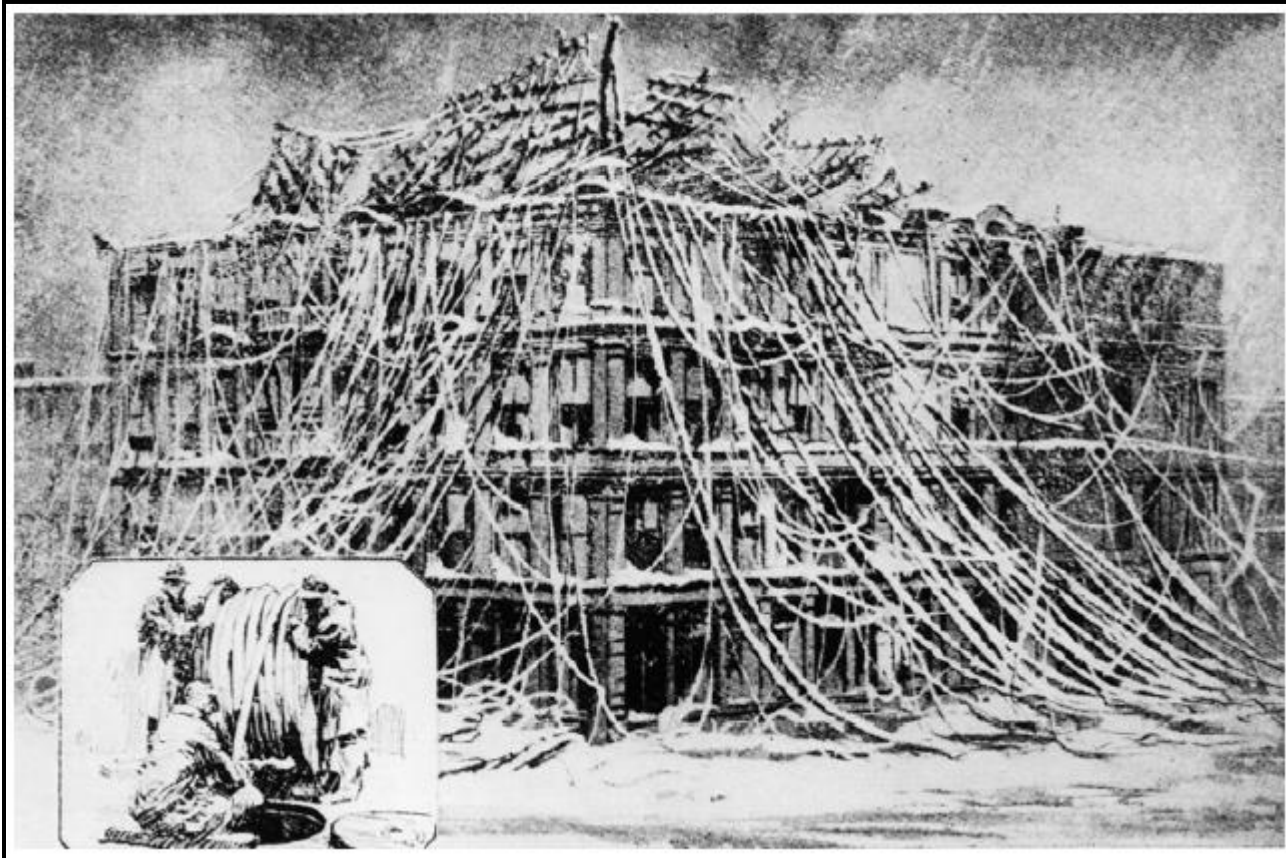


Another Example of Computer Revolution



<i>Original IBM PC (1981)</i>	<i>MICAZ Mote (2005)</i>
4.77 MHz	4 MHz
16-256 KB RAM	128 KB RAM
160 KB Floppies	512 KB Flash
~ \$6K (today)	~ \$35
~ 64 W	~14 mW
25 lb, 19.5 x 5.5 x 16 inch	0.5 oz, 2.25 x 1.25 x 0.25 inch

Another Revolution to Reach Here: **Wireless Revolution**



Boston central telephone station at 40 Pearl Street after the blizzard of 1881

Technology Projection Challenge for Energy: Communication/Computation

Bluetooth Technology	1999	2004
Communication	(150nJ/bit)	(5nJ/bit)
	1.5mW*	50uW
Computation		~ 190 MOPS
		(5pJ/OP)

Assume: 10kbit/sec. Radio, 10 m range.

Large cost of communications relative to computation continues

Bottom Line ...

- ▶ By **Computer Systems** we often mean
 - ▶ General-purpose, digital computers that interact with humans and possibly with each other.
- ▶ Change in Trend using **Embedded Systems**
 - ▶ More special purpose-computers
 - ▶ More scenarios without humans in the loop
 - ▶ More environments where computing is not the explicit / main goal.

Bottom Line ...

▶ **Computer Systems**

- ▶ Interactive (HCI)
- ▶ Discrete Time not counted
- ▶ Stand-alone / Networked
- ▶ Isolated Environment

▶ **Embedded Systems**

- ▶ Reactive (Event-driven)
- ▶ **Timeliness** is important to critical
- ▶ Computing Component may be controlled by ***external system***
- ▶ ***Alien Environment*** (may affect function)

Implications (*of being embedded*)

- ▶ **Low cost hardware**

- ▶ Software Development cost (one-time) vs. Per-shipped-item cost (recurring)

- ▶ **Very Limited Storage Space**

- ▶ Convoluted Algorithms and Aggressive Optimization
- ▶ Space complexity !!

- ▶ **Development Environment is different from execution environment**

- ▶ Can't do all time test-bed implementation, rather depending on simulation
- ▶ Constrained process; innovative techniques?

Implications (*of being embedded*)

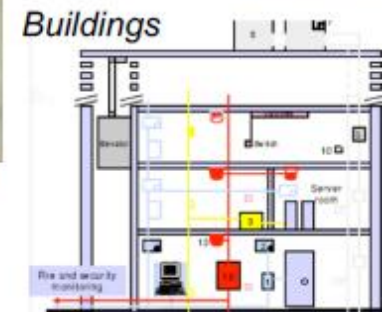
- ▶ **Long and un-interrupted running times**
 - ▶ Always running system !!
 - ▶ Can't reset flight control system or cardiac pacemaker at operation
- ▶ **Alien (Uncontrolled) Environments**
 - ▶ Harsh and human un-attendant scenario
 - ▶ Increase reliability constraints
- ▶ **Timeliness expectations**
 - ▶ Time complexity !!
 - ▶ Specific approaches for design, development and testing

Cyber Physical System (CPS)

- ▶ Cyber-physical systems (CPS) are engineered systems that are built from, and depend upon, the **seamless integration** of computation and physical components.
- ▶ Advances in CPS will **enable** capability, adaptability, scalability, resiliency, safety, security, and usability that will expand the horizons of these critical systems.
- ▶ CPS technologies are **transforming the way people interact** with engineered systems, just as the Internet has transformed the way people interact with information.

Cyber Physical System (CPS)

- ▶ Cyber + Physical
- ▶ Computation + Dynamics + Communication
- ▶ Security + Safety



Application Domains of CPS

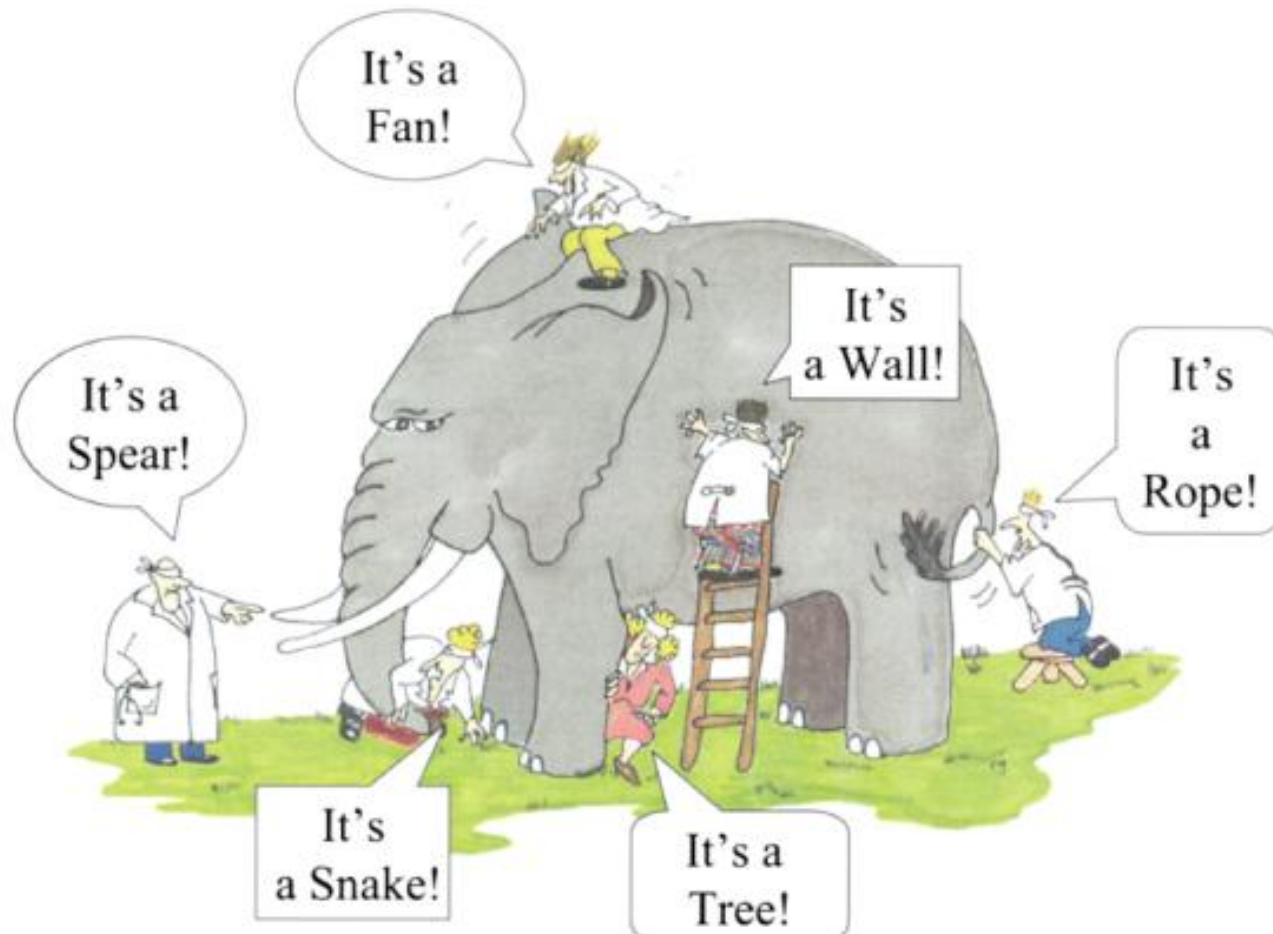
- ▶ Agriculture
- ▶ Aeronautics
- ▶ Building Design
- ▶ Civil Infrastructure
- ▶ Energy
- ▶ Environmental Quality
- ▶ Healthcare
- ▶ Personalized Medicine
- ▶ Industrial Manufacturing
- ▶ Transportation

Contradictions in CPS Decision Making

- ▶ **Adaptability** vs. **Repeatability**
- ▶ **High Connectivity** vs. **Security and Privacy**
- ▶ **High Performance** vs. **Low Energy**
- ▶ **Asynchrony** vs. **Coordination/Cooperation**
- ▶ **Scalability** vs. **Reliability and Predictability**
- ▶ **Laws and Regulations** vs. **Technical Possibilities**
- ▶ **Economics of Computing Scale (i.e., Cloud Computing)** vs. **Locality (i.e., Fog Computing)**
- ▶ **Open** vs. **Proprietary**
- ▶ **Algorithms** vs. **Dynamics**

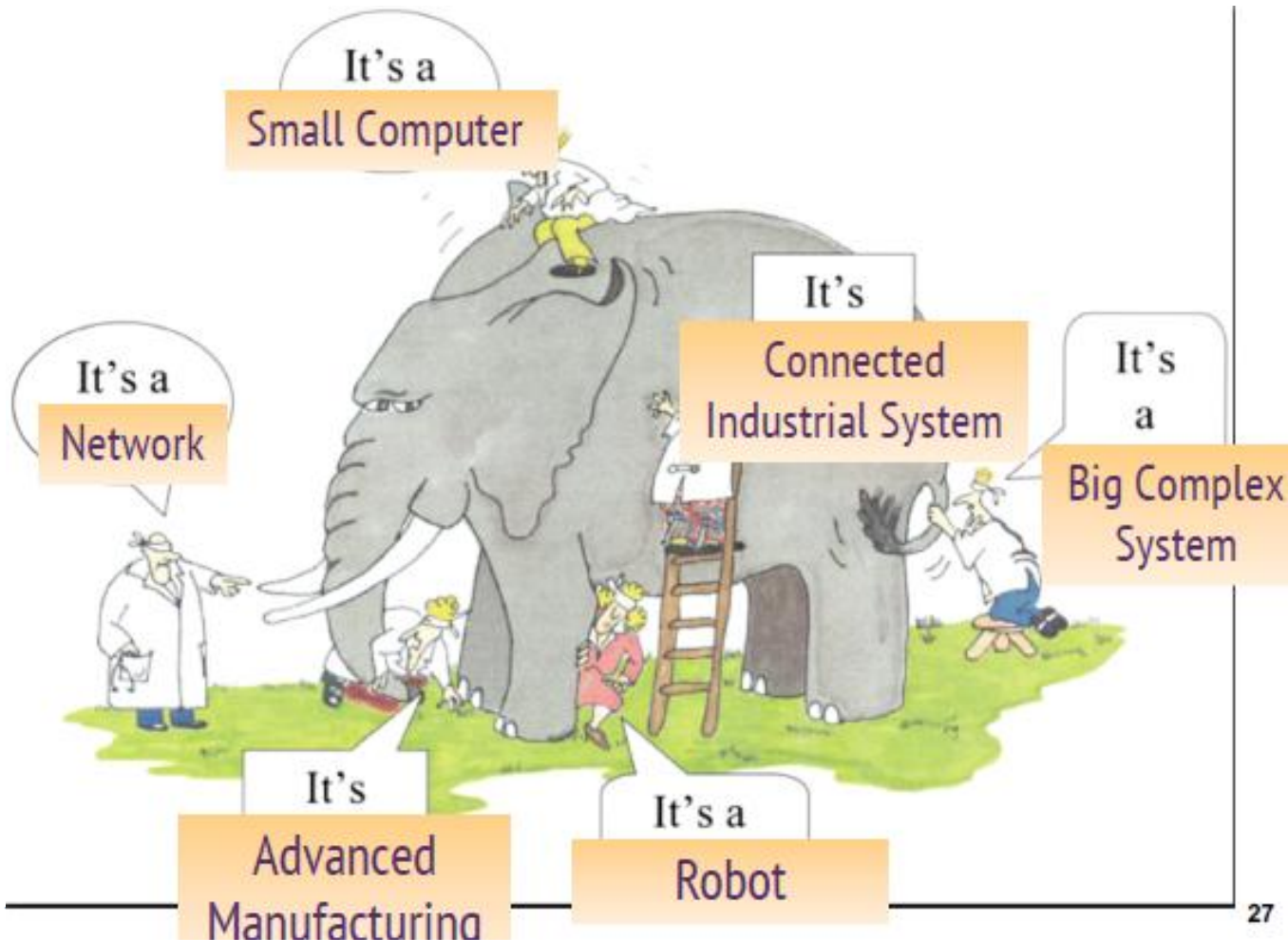
So CPS is a Multidisciplinary Area !!!

► Challenges of Working in CPS



So CPS is a Multidisciplinary Area !!!

► Challenges of Working in CPS



CPS System: STARMAC Quadrotor Aircraft

- ▶ **Introductory Video:**

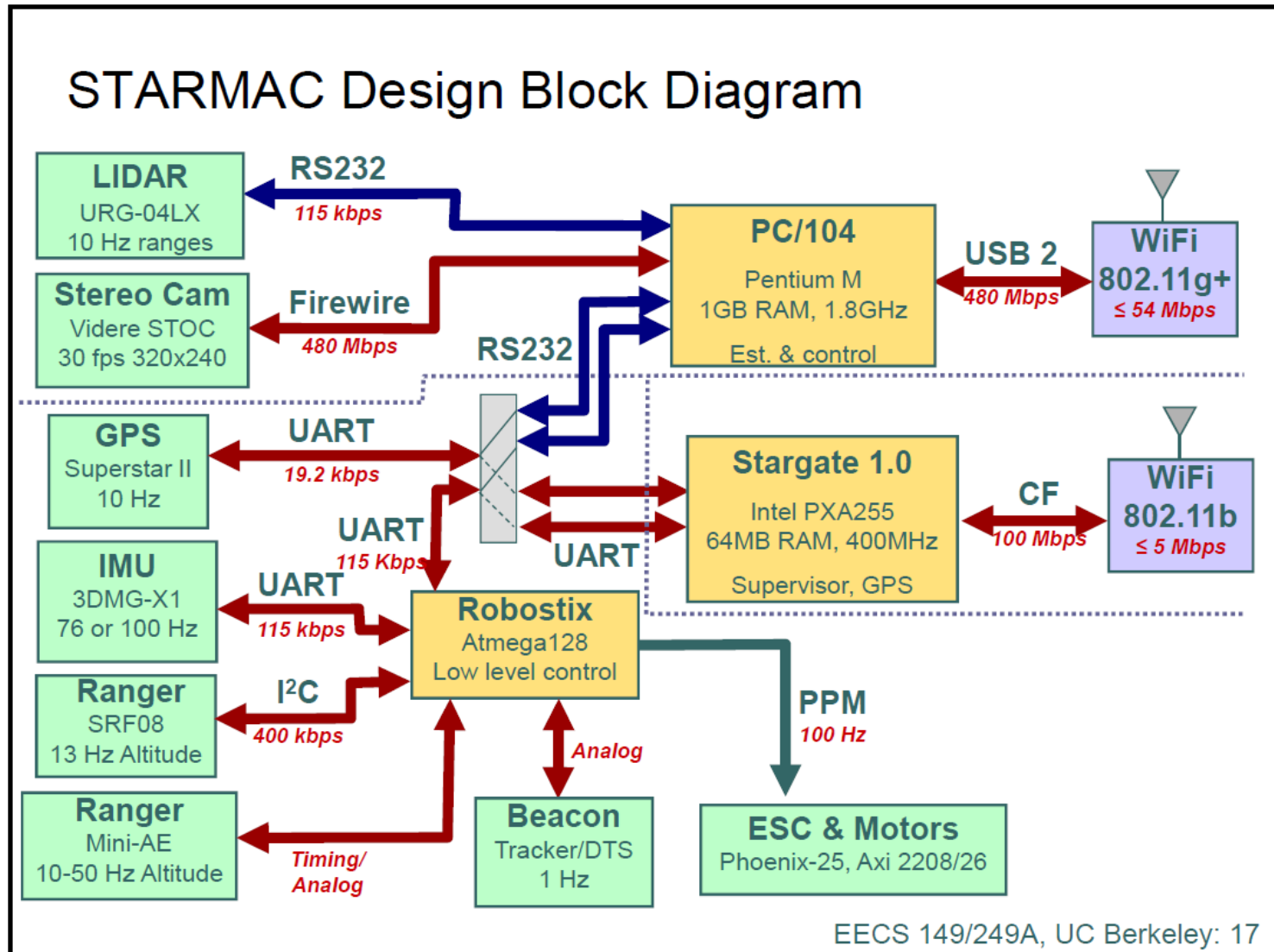
<http://www.youtube.com/watch?v=rJ9r2orcaYo>

- ▶ **Back-Flip Manuever:**

<http://www.youtube.com/watch?v=iD3QgGpzzlM>



CPS System: STARMAC Quadrotor Aircraft

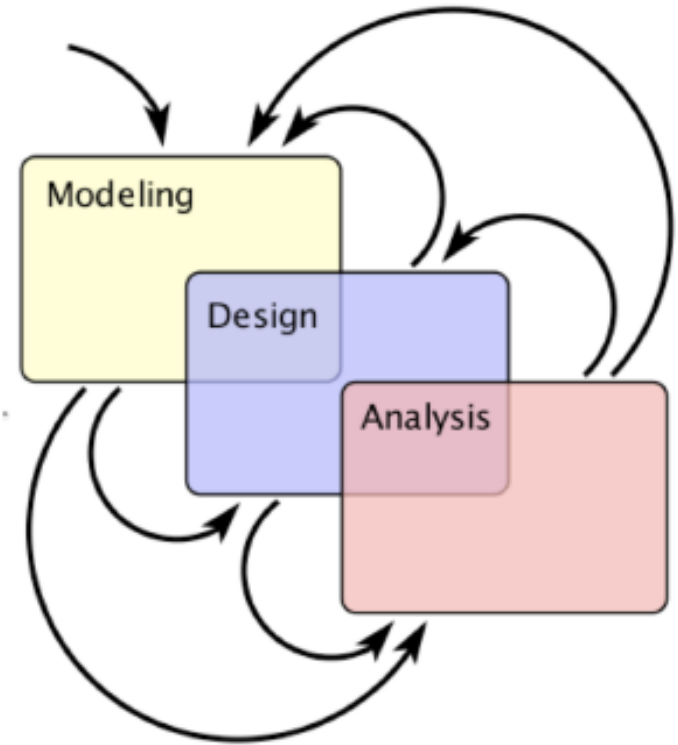


What is this CPS About?

- ▶ A scientific structured approach for **designing and implementing embedded systems**
- ▶ It is not just **hacking** and **implementing**
- ▶ Focus on model-based system design, on **embedded** hardware and software

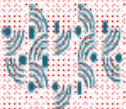
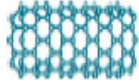
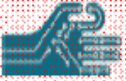
CPS Model, Design and Analysis

- ▶ **Modeling** is the process of gaining a deeper understanding of a system through imitation. Models specify **what** a system does.
- ▶ **Design** is the structured creation of artifacts. It specifies **how** a system does, what it does. This includes optimization.
- ▶ **Analysis** is the process of gaining a deeper understanding of a system through dissection. It specifies **why** a system does, what it does (or fails to do what a model says it should do).



CPS Relevance: McKinsey's Disruptive Technologies

Twelve potentially economically disruptive technologies

	Mobile Internet	Increasingly inexpensive and capable mobile computing devices and Internet connectivity		Next-generation genomics	Fast, low-cost gene sequencing, advanced big data analytics, and synthetic biology ("writing" DNA)
	Automation of knowledge work	Intelligent software systems that can perform knowledge work tasks involving unstructured commands and subtle judgments		Energy storage	Devices or systems that store energy for later use, including batteries
	The Internet of Things	Networks of low-cost sensors and actuators for data collection, monitoring, decision making, and process optimization		3D printing	Additive manufacturing techniques to create objects by printing layers of material based on digital models
	Cloud technology	Use of computer hardware and software resources delivered over a network or the Internet, often as a service		Advanced materials	Materials designed to have superior characteristics (e.g., strength, weight, conductivity) or functionality
	Advanced robotics	Increasingly capable robots with enhanced senses, dexterity, and intelligence used to automate tasks or augment humans		Advanced oil and gas exploration and recovery	Exploration and recovery techniques that make extraction of unconventional oil and gas economical
	Autonomous and near-autonomous vehicles	Vehicles that can navigate and operate with reduced or no human intervention		Renewable energy	Generation of electricity from renewable sources with reduced harmful climate impact

Economic Potential of Technologies

	The Internet of Things	300% Increase in connected machine-to-machine devices over past 5 years 80–90% Price decline in MEMS (microelectromechanical systems) sensors in past 5 years	1 trillion Things that could be connected to the Internet across industries such as manufacturing, health care, and mining 100 million Global machine to machine (M2M) device connections across sectors like transportation, security, health care, and utilities	\$36 trillion Operating costs of key affected industries (manufacturing, health care, and mining)
	Cloud technology	18 months Time to double server performance per dollar 3x Monthly cost of owning a server vs. renting in the cloud	2 billion Global users of cloud-based email services like Gmail, Yahoo, and Hotmail 80% North American institutions hosting or planning to host critical applications on the cloud	\$1.7 trillion GDP related to the Internet \$3 trillion Enterprise IT spend
	Advanced robotics	75–85% Lower price for Baxter ⁹ than a typical industrial robot 170% Growth in sales of industrial robots, 2009–11	320 million Manufacturing workers, 12% of global workforce 250 million Annual major surgeries	\$6 trillion Manufacturing worker employment costs, 19% of global employment costs \$2–3 trillion Cost of major surgeries
	Autonomous and near-autonomous vehicles	7 Miles driven by top-performing driverless car in 2004 DARPA Grand Challenge along a 150-mile route 1,540 Miles cumulatively driven by cars competing in 2005 Grand Challenge 300,000+ Miles driven by Google's autonomous cars with only 1 accident (which was human-caused)	1 billion Cars and trucks globally 450,000 Civilian, military, and general aviation aircraft in the world	\$4 trillion Automobile industry revenue \$155 billion Revenue from sales of civilian, military, and general aviation aircraft

Thank You !!

